

Laboratory work 1. ERD diagram

Course: Databases | System: International Airport

1. Entities

The following 10 entities were identified from the system description:

#	Entity	Meaning
1	AIRPORT	Stores airports and when the record was created/updated.
2	AIRLINE	Airline companies that operate flights.
3	FLIGHT	Scheduled and actual flight information.
4	PASSENGER	Passengers associated with bookings.
5	BOOKING	A ticket booking of one passenger for one flight.
6	BOOKING_HISTORY	Separate table that stores every change made to a booking.
7	BOARDING_PASS	Boarding pass issued for a booking.
8	BAGGAGE	Baggage registered during booking.
9	BAGGAGE_CHECK	Airport check of the baggage that belongs to a booking.
10	SECURITY_CHECK	Airport security check passed by a passenger.

2. Attributes, keys and data types

PK = primary key, FK = foreign key. All timestamps are TIMESTAMP; surrogate keys are INT with AUTO_INCREMENT.

AIRPORT

Key	Attribute	Data type	Constraints	Description
PK	airport_id	INT	NOT NULL, AUTO_INCREMENT	Unique airport identifier
	airport_name	VARCHAR(100)	NOT NULL	Name of the airport
	country	VARCHAR(60)	NOT NULL	Country
	state	VARCHAR(60)	NULL	State / region
	city	VARCHAR(60)	NOT NULL	City
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

AIRLINE

Key	Attribute	Data type	Constraints	Description
PK	airline_id	INT	NOT NULL, AUTO_INCREMENT	Unique airline identifier
	airline_code	CHAR(3)	NOT NULL, UNIQUE	Airline code (e.g. KC)
	name	VARCHAR(100)	NOT NULL	Airline name
	country	VARCHAR(60)	NOT NULL	Country of the airline
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

FLIGHT

Key	Attribute	Data type	Constraints	Description
PK	flight_id	INT	NOT NULL, AUTO_INCREMENT	Unique flight identifier
FK	airline_id	INT	NOT NULL	References AIRLINE
FK	departure_airport_id	INT	NOT NULL	References AIRPORT (departure)
FK	arrival_airport_id	INT	NOT NULL	References AIRPORT (arrival)
	departing_gate	VARCHAR(10)	NULL	Departure gate
	arriving_gate	VARCHAR(10)	NULL	Arrival gate
	scheduled_departure	TIMESTAMP	NOT NULL	Scheduled departure time
	scheduled_arrival	TIMESTAMP	NOT NULL	Scheduled arrival time
	actual_departure	TIMESTAMP	NULL	Actual departure time
	actual_arrival	TIMESTAMP	NULL	Actual arrival time
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

PASSENGER

Key	Attribute	Data type	Constraints	Description
PK	passenger_id	INT	NOT NULL, AUTO_INCREMENT	Unique passenger identifier
	first_name	VARCHAR(50)	NOT NULL	First name
	last_name	VARCHAR(50)	NOT NULL	Last name
	gender	CHAR(1)	NULL	Gender (M/F)
	date_of_birth	DATE	NOT NULL	Date of birth
	country_of_citizenship	VARCHAR(60)	NOT NULL	Citizenship
	country_of_residence	VARCHAR(60)	NOT NULL	Country of residence
	passport_number	VARCHAR(20)	NOT NULL, UNIQUE	Passport number
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

BOOKING

Key	Attribute	Data type	Constraints	Description
PK	booking_id	INT	NOT NULL, AUTO_INCREMENT	Unique booking identifier
FK	flight_id	INT	NOT NULL	References FLIGHT
FK	passenger_id	INT	NOT NULL	References PASSENGER
	status	VARCHAR(20)	NOT NULL	Booking status (booked, paid, cancelled...)
	booking_platform	VARCHAR(30)	NULL	Website, mobile app, agency...
	ticket_price	DECIMAL(10,2)	NOT NULL	Ticket price
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

BOOKING_HISTORY

Key	Attribute	Data type	Constraints	Description
PK	history_id	INT	NOT NULL, AUTO_INCREMENT	Unique history record identifier
FK	booking_id	INT	NOT NULL	References BOOKING
	old_status	VARCHAR(20)	NULL	Status before the change
	new_status	VARCHAR(20)	NULL	Status after the change
	old_ticket_price	DECIMAL(10,2)	NULL	Price before the change
	new_ticket_price	DECIMAL(10,2)	NULL	Price after the change
	changed_at	TIMESTAMP	NOT NULL	Time of the change

BOARDING_PASS

Key	Attribute	Data type	Constraints	Description
PK	boarding_pass_id	INT	NOT NULL, AUTO_INCREMENT	Unique boarding pass identifier
FK	booking_id	INT	NOT NULL, UNIQUE	References BOOKING (one pass per booking)
	seat	VARCHAR(5)	NOT NULL	Seat number (e.g. 12A)
	boarding_time	TIMESTAMP	NOT NULL	Boarding time
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

BAGGAGE

Key	Attribute	Data type	Constraints	Description
PK	baggage_id	INT	NOT NULL, AUTO_INCREMENT	Unique baggage identifier
FK	booking_id	INT	NOT NULL	References BOOKING
	weight_kg	DECIMAL(5,2)	NOT NULL	Weight in kg
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

BAGGAGE_CHECK

Key	Attribute	Data type	Constraints	Description
PK	baggage_check_id	INT	NOT NULL, AUTO_INCREMENT	Unique baggage check identifier
FK	booking_id	INT	NOT NULL	References BOOKING
FK	passenger_id	INT	NOT NULL	References PASSENGER
	check_result	VARCHAR(20)	NOT NULL	Result (passed, failed)
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

SECURITY_CHECK

Key	Attribute	Data type	Constraints	Description
PK	security_check_id	INT	NOT NULL, AUTO_INCREMENT	Unique security check identifier
FK	passenger_id	INT	NOT NULL	References PASSENGER
	check_result	VARCHAR(20)	NOT NULL	Result (passed, failed)
	created_at	TIMESTAMP	NOT NULL	Record creation time
	updated_at	TIMESTAMP	NOT NULL	Last update time

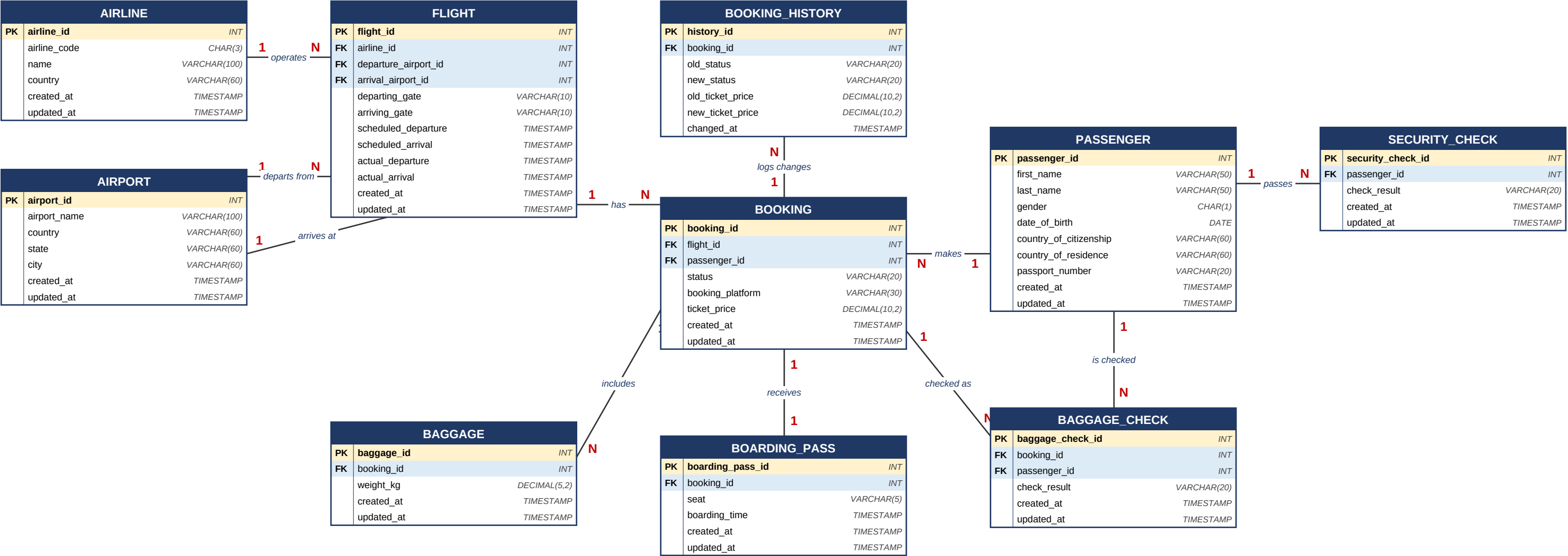
3. Relationships, cardinality and participation

#	Entity A	Relationship	Entity B	Cardinality	Participation
1	AIRLINE	operates	FLIGHT	1 : N	AIRLINE partial, FLIGHT total
2	AIRPORT	is departure airport of	FLIGHT	1 : N	AIRPORT partial, FLIGHT total
3	AIRPORT	is arrival airport of	FLIGHT	1 : N	AIRPORT partial, FLIGHT total
4	FLIGHT	has	BOOKING	1 : N	FLIGHT partial, BOOKING total
5	PASSENGER	makes	BOOKING	1 : N	PASSENGER partial, BOOKING total
6	BOOKING	is logged in	BOOKING_HISTORY	1 : N	BOOKING partial, HISTORY total
7	BOOKING	receives	BOARDING_PASS	1 : 1	BOOKING partial, PASS total
8	BOOKING	includes	BAGGAGE	1 : N	BOOKING partial (baggage optional), BAGGAGE total
9	BOOKING	is covered by	BAGGAGE_CHECK	1 : N	BOOKING partial, CHECK total
10	PASSENGER	is subject of	BAGGAGE_CHECK	1 : N	PASSENGER partial, CHECK total
11	PASSENGER	passes	SECURITY_CHECK	1 : N	PASSENGER partial, CHECK total

The ER diagram is on the next page.

ER Diagram - International Airport Database

Notation: entity boxes with attributes (PK/FK), lines show relationships with cardinality 1 and N.



Legend

- | | |
|-------------------|--|
| ENTITY | Entity (table); its name is in the header, attributes are listed inside the box |
| PK id | Primary key (unique identifier of a row), shown in bold on yellow |
| FK id | Foreign key (references the primary key of another entity), shown on blue |
| TYPE | Data type of the attribute is written in italics at the right side of each row |
| 1 N | Relationship line: 1 = one record, N = many records (one-to-many); label = verb |
| | 1 - 1 on both ends means one-to-one (BOOKING - BOARDING_PASS) |
| | Participation (total/partial) of every relationship is given in the table in section 3 |

4. Description of the solution

The model describes how an airport manages flights, passenger bookings, baggage and checks. Every entity has its own surrogate primary key, and every relationship is implemented with a foreign key in the 'many' side table.

Meaning of entities and relationships

- **AIRLINE - FLIGHT (1:N)**. One airline operates many flights; each flight belongs to exactly one airline (FLIGHT.airline_id).
- **AIRPORT - FLIGHT (1:N, twice)**. An airport can be the departure airport of many flights and the arrival airport of many flights. FLIGHT therefore has two foreign keys to AIRPORT: departure_airport_id and arrival_airport_id.
- **FLIGHT - BOOKING (1:N)**. A flight has many bookings; each booking is made for one flight.
- **PASSENGER - BOOKING (1:N)**. A passenger can make many bookings; each booking belongs to one passenger.
- **BOOKING - BOOKING_HISTORY (1:N)**. Any change of a booking (status, price) is written as a new row into a separate history table.
- **BOOKING - BOARDING_PASS (1:1)**. One boarding pass is issued for one booking (UNIQUE foreign key booking_id).
- **BOOKING - BAGGAGE (1:N)**. A passenger may register several pieces of baggage or none.
- **BOOKING / PASSENGER - BAGGAGE_CHECK (1:N)**. Baggage that belongs to a booking is checked in the airport; the check row stores the booking and the passenger.
- **PASSENGER - SECURITY_CHECK (1:N)**. Each passenger passes the airport security check; the result is stored in SECURITY_CHECK.

Design decisions

- Many-to-many relationships are not needed: the link between passengers and flights is resolved by the BOOKING entity.
- Common attributes created_at and updated_at are stored in all main tables to track when a record was created and changed.
- Actual departure/arrival times and gates are nullable, because they are unknown until the flight takes place.
- passport_number and airline_code are UNIQUE; foreign keys guarantee referential integrity between tables.
- The tables are in third normal form: each non-key attribute depends only on the primary key of its table.